

Naga Venkata Satya Sai Krishna Munukutla

AI Product Lead · Sovereign AI Architect

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Work permit: Residencia y Trabajo — valid until October 2028

PROFILE

Ten years in technology — starting in IoT and embedded systems, moving through cloud infrastructure, and now focused on AI product development. At Sortis I own the AI vertical: defining what gets built, designing the architecture, and delivering it to production. My work centres on sovereign AI — systems that run entirely on private infrastructure, with no data sent to external APIs. The five products I have shipped under the Veltrix name (Veritas, Sage, Beacon, FORGE, and NEXUS) all follow that principle. I work closely with Zadara Cloud Solutions on the infrastructure layer and have co-authored published technical content on sovereign AI architecture.

EXPERIENCE

AI Product Lead — Sortis · Zadara Cloud Solutions 2024 – Present

Responsible for the AI vertical roadmap — both short-term deliverables and longer-term direction — building internal and external AI products in partnership with Zadara.

- Veltrix Veritas v2.0 — a Document Intelligence Platform that takes a single technical document and produces a complete sales intelligence package: competitive analysis via self-hosted SearXNG, winning arguments with evidence citations, audience profiles across 16 industry verticals, and multi-channel campaign plans. Runs on private vLLM (Qwen/Qwen2.5-VL-7B-Instruct-AWQ), ChromaDB, and LangChain. Zero data leaves the network.
- Veltrix Sage — organisational knowledge query engine. FastAPI backend, Streamlit UI, ChromaDB vector store, GDPR-compliant, bilingual ES/EN. Production v1.0 deployed on-premise.
- Veltrix Beacon — Microsoft 365 knowledge discovery with cross-platform PC agents. Same private-infrastructure principle; full production deployment.
- NEXUS — end-to-end sovereign AI service pack for enterprise clients. Includes the FORGE embedded AI team-as-a-service offering.
- Co-authored a technical article on sovereign AI architecture with Zadara's engineering team, published on the Zadara blog.
- Led architecture and pricing for a €259K public procurement bid (SUMA Gestión Tributaria — generative AI virtual assistant), including compliance mapping across contract documents.

Senior Cloud & AI Engineer — Sortis Telecommunications · ST Engineering iDirect Jun 2021 – 2024

Customer-facing cloud engineering for satellite network clients. The work was a mix of infrastructure design, Python automation, and direct client engagement across Europe, the US, and India.

- Led an AI digitalization initiative — built pipelines on AWS to replace manual satellite support workflows with predictive monitoring and automated triage.
- Designed serverless and containerised infrastructure (Lambda, ECS/Fargate, S3, RDS, CloudWatch) for real-time network diagnostics across client deployments.
- Built time-series forecasting and anomaly detection on satellite telemetry data; reduced reactive incidents by catching signal degradation before it reached customers.
- Wrote secure REST APIs connecting customer systems to internal support tooling. Also handled field training for international client teams.

- SLM (Server Lifecycle Manager) — built an internal full-stack IT platform on Vue 3, Django 5, PostgreSQL, Redis/Celery, and Nginx; currently in production at slm.sortis.es.
- NexSignal — developed an Android app for cellular signal intelligence and dual-SIM carrier switching targeting Orange Spain; produced the full pitch deck and technical alignment documentation.

Data Engineer Intern — ADEX Corporation

Nov 2020 – Jun 2021

- Built Python data pipelines for real-time industrial sensor data with automated alarm generation, integrated into a proprietary control optimisation platform.
- Developed SQL-backed dashboards for power plant performance monitoring and predictive maintenance.

Hardware & Embedded Engineer — MSR Technologies, India

2016 – 2017

- Designed sensor-fusion systems on Arduino and Raspberry Pi for home automation and smart parking. Wrote control algorithms in Python and C.

TECHNICAL SKILLS

AI & LLM: LLM, LangChain, ChromaDB, RAG pipelines, SearXNG, Streamlit, spaCy, Scikit-learn, ML concepts

Cloud: AWS — Lambda, ECS/Fargate, EC2, S3, RDS, IAM, VPC, CloudWatch, Secrets Manager, API Gateway

Languages: Python (advanced), SQL (advanced), Bash, JavaScript (working knowledge)

Data: Snowflake, Apache Iceberg, Airflow (MWAA), MongoDB, Pandas

DevOps & IaC: Terraform, Docker, Azure DevOps, Jenkins, Git, CI/CD pipelines

Full-stack: Vue 3, Django 5 / DRF, FastAPI, PostgreSQL, Redis, Nginx

App development: Android (NexSignal — dual-SIM signal intelligence)

EDUCATION

MSc Telecommunication Engineering — Universidad de Jaén, Spain

2018 – 2021

Thesis: Arduino-based photovoltaic energy harvesting with MPPT control (500+ hours, MATLAB/Simulink validation)

BTech Electronics & Communications Engineering — Andhra University, India

2014 – 2018

Hackathons with Cisco and Smart India Hackathon; presented startup ideas at APXLR8 accelerator

CERTIFICATIONS

AWS Cloud Practitioner (Amazon Web Services)

Azure Cloud Concepts (Microsoft)

Cybersecurity Essentials (Cisco)

In progress: AWS Solutions Architect Associate

LANGUAGES

Spanish (Fluent) · English (C1 — Fluent) · Telugu (Native) · Hindi (Fluent)